HDOC 62

MERGING TWO SORTED ARRAYS USING POINTERS

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Problem Statement

Write a C program that reads the sizes and elements of two sorted variable-length integer arrays, arr1[] and arr2[], containing their elements in ascending order. Pass both arrays to a function named merge_sorted_arrays() using their addresses. Use pointers to pass the arrays between the calling function and merge_sorted_arrays(). The function merge_sorted_arrays() must:

- Merge arr1[] and arr2[] into a third array, merged[].
- Maintain the ascending order of the elements in merged[].
- Preserve duplicate elements, if any.
- Display the elements of merged[] separated by spaces.
- Use the two-pointer technique to merge the arrays efficiently.

Algorithm

1. Start.

2. Read the size n1 and elements of the first sorted array arr1[].

3. Read the size n2 and elements of the second sorted array arr2[].

4. Create merged[] of size n1 + n2.

5. Pass arr1, arr2 and merged to merge_sorted_arrays() using their addresses.

6. Initialize i = 0, j = 0 and k = 0.

7. While both arrays contain unprocessed elements, compare *(arr1 + i) and *(arr2 + j). Copy the smaller element into *(merged + k), then advance the corresponding pointer index and k.

8. Copy any remaining elements of arr1[] into merged[].

9. Copy any remaining elements of arr2[] into merged[].

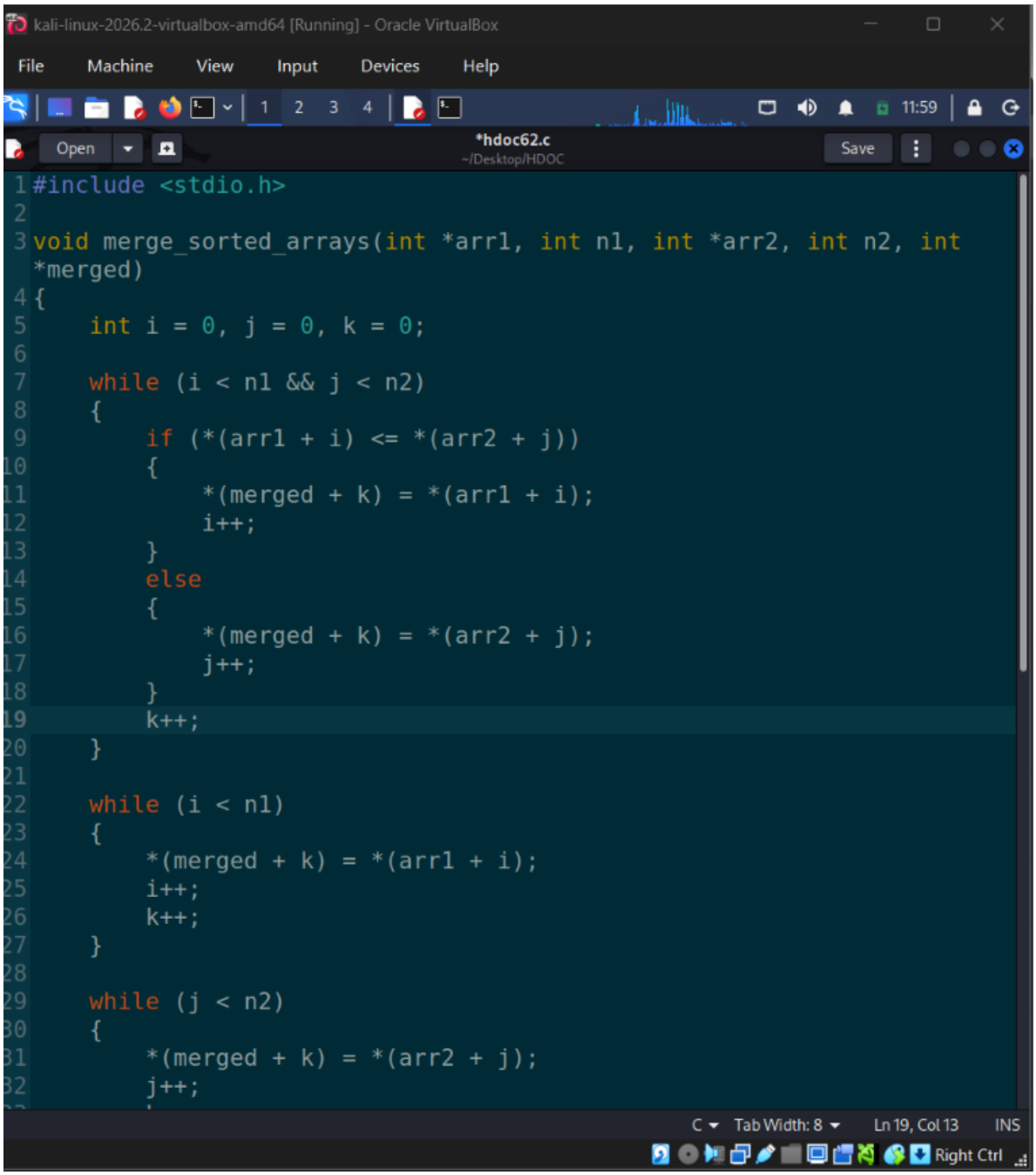
10. Display all elements of merged[] separated by spaces.

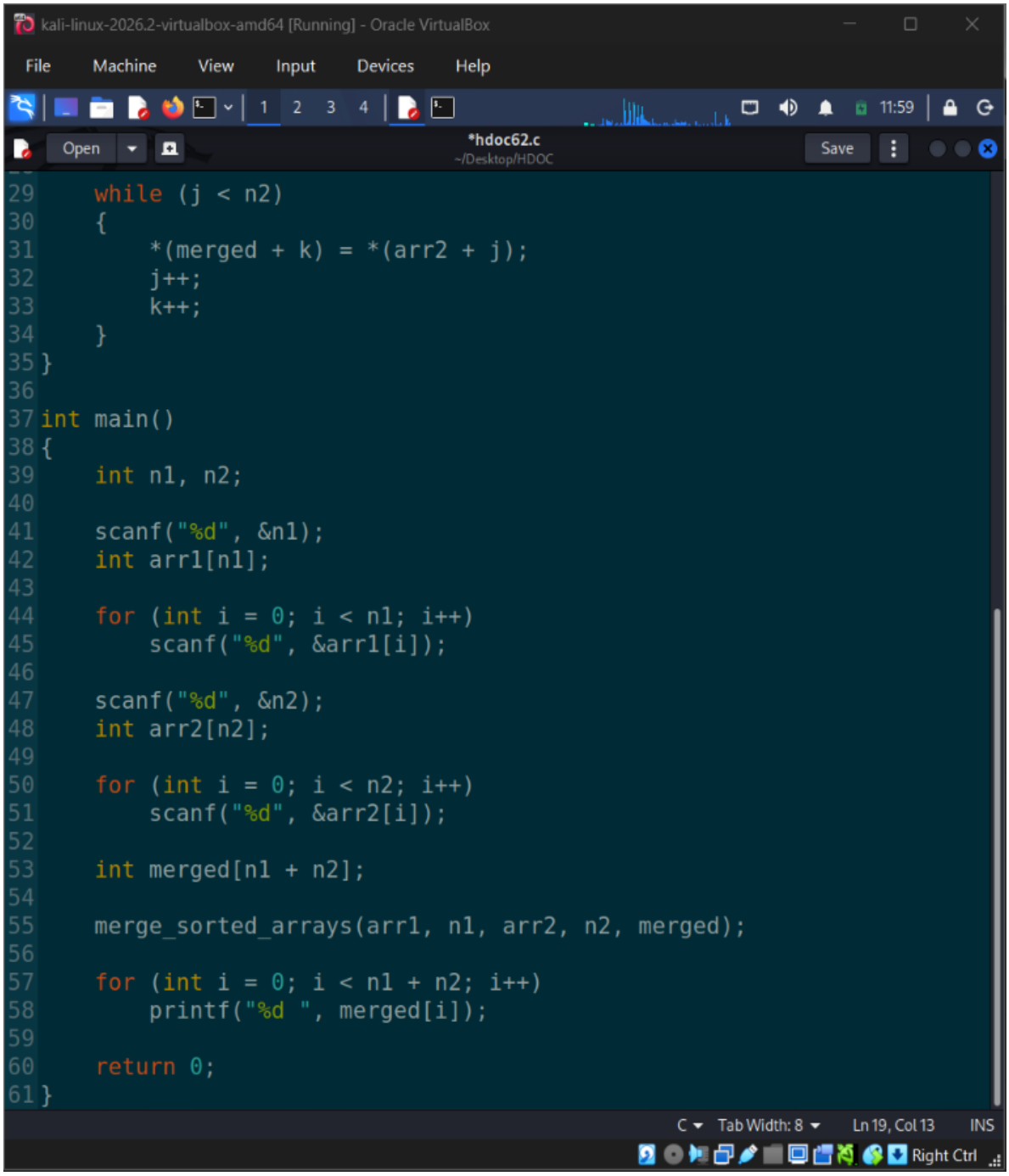
11. Stop.

Test Case Table

S. No.	arr1[]	arr2[]	Expected Output	Actual Output	Result
1	1 3 5 7 9	2 3 6 8	1 2 3 3 5 6 7 8 9	1 2 3 3 5 6 7 8 9	Pass
2	1 2 2 5	2 2 3 5 6	1 2 2 2 2 3 5 5 6	1 2 2 2 2 3 5 5 6	Pass
3	1 4 10	2 3 5 6 7 8	1 2 3 4 5 6 7 8 10	1 2 3 4 5 6 7 8 10	Pass
4	-10 -5 0 5	-8 -3 2 7	-10 -8 -5 -3 0 2 5 7	-10 -8 -5 -3 0 2 5 7	Pass
5	5	10	5 10	5 10	Pass

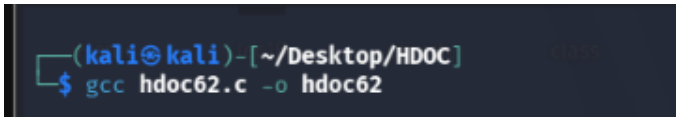
C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
*hdoc62.c
~/Desktop/HDOC
Save
#include <stdio.h>
void merge_sorted_arrays(int *arr1, int n1, int *arr2, int n2, int *merged)
{
int i = 0, j = 0, k = 0;
while (i < n1 && j < n2)
{
if (*(arr1 + i) <= *(arr2 + j))
{
*(merged + k) = *(arr1 + i);
i++;
}
else
{
*(merged + k) = *(arr2 + j);
j++;
}
k++;
}
while (i < n1)
{
*(merged + k) = *(arr1 + i);
i++;
k++;
}
while (j < n2)
{
*(merged + k) = *(arr2 + j);
j++;
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Compilation and Execution

Compilation command: gcc hdoc62.c -o hdoc62



Execution command: ./hdoc62

Execution Screenshot:

